

Searchable.City: An Open-Vocabulary Semantic Atlas

Sean Hardesty Lewis

Cornell Tech

New York, NY, USA

shl225@cornell.edu



Figure 1: *Searchable.City* localizes visual patterns across city-scale street-level imagery. A query for graffiti (left) highlights dense corridor clusters, while bridges (right) traces New York City’s perimeter and crossings. Each map view is paired with representative retrieved images (3 × 3).

Abstract

Maps excel at locations and are famously poor at surfaces: the fire escapes, awnings, scaffolds, shade, murals, and textures that make cities legible to residents rarely exist as database fields. We present *Searchable.City*, a pipeline that treats the city as an image corpus rather than only a registry of points of interest. By running a vision–language model (VLM) over millions of street-level images and indexing the resulting descriptions, we construct an *open-vocabulary semantic atlas*: a map that can be searched for meanings (“Chinese”, “gothic”, “construction”) instead of addresses. We show how this translation, from city to caption to queryable field, reveals cultural gradients, architectural memory, and infrastructural churn, while also clarifying the limits of the street-view vantage. Open-vocabulary mapping is not only a technical interface but also a cultural instrument: it redraws neighborhood boundaries according to what the machine can name, and what it cannot see.

CCS Concepts

• **Information systems** → **Geographic information systems**; • **Computing methodologies** → **Computer vision**.

ACM Reference Format:

Sean Hardesty Lewis. 2026. Searchable.City: An Open-Vocabulary Semantic Atlas. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH Emerging Technologies '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3799826.3812433>



This work is licensed under a Creative Commons Attribution 4.0 International License. *SIGGRAPH Emerging Technologies '26*, Los Angeles, CA, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2549-4/26/07
<https://doi.org/10.1145/3799826.3812433>

1 Introduction

Current GIS systems represent the city as vectors and points, but they miss the semantic texture of urban life. A digital map knows where the pharmacy is, but rarely where the fire escape begins. We present *Searchable.City*, an interactive system that asks the city to describe itself. By processing millions of street-view images with a vision–language model (VLM), we build a queryable spatial index that lets users search the physical world for arbitrary visual concepts, such as “art deco” and “trash piles”. This shifts the map from a navigational tool into a semantic search engine.

Prior work in urban computing has used street-view imagery mostly for scalar prediction, such as estimating perceived safety [Naik et al. 2014]. In parallel, open-vocabulary vision models such as CLIP [Radford et al. 2021] and SAM [Kirillov et al. 2023] enable recognition without a fixed label set. We bridge these strands by using instruction-tuned VLMs [Vasu et al. 2024] to generate a broad, open-ended lexicon of the streetscape, moving beyond pre-defined classes to capture the long tail of urban details.

2 Methodological Innovations

The core innovation of *Searchable.City* is a shift in the mapping primitive: mapping language onto geography, rather than only coordinates onto images. The pipeline begins by ingesting a large corpus of street-view imagery while retaining geospatial coordinates and camera heading. To handle the dense overlap of urban photography, we discretize the streetscape into directional “vantage cells” (Figure 2) so that queries align with what is visible from the street, not only with a top-down location.

For each vantage cell, a VLM generates a detailed descriptive caption. We decompose this caption into a multi-tag representation, producing a lexicon of thousands of descriptors. Unlike fixed-class

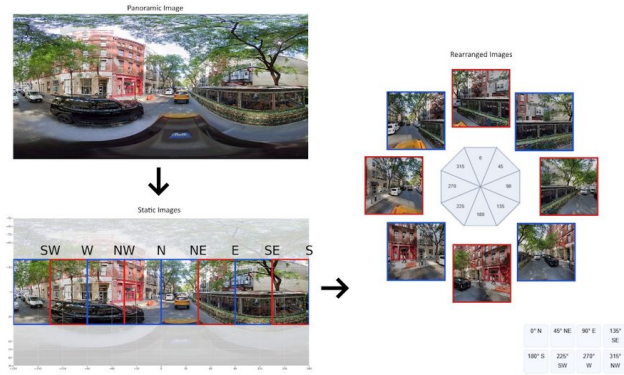


Figure 2: Vantage Cell Discretization: The streetscape is divided into directional slices based on available street-view metadata, creating spatial buckets for query aggregation.

segmentation, this lets the system index ad-hoc concepts like “temporary scaffolding” or “art deco.” Tags are then spatially aggregated into a probabilistic index. The interface works like a search engine: a user inputs a string, and the system returns a semantic field showing where that visual description clusters across the city.

The system is intentionally evidence-preserving. Each query returns not only an aggregated spatial layer but also representative retrieved images, allowing users to inspect the visual basis of the result. This is important because open-vocabulary categories are rarely clean object classes. A search for “gothic,” for example, may retrieve arches, stone ornament, church facades, or dark vertical massing; a search for “construction” may retrieve scaffolds, sidewalk sheds, cranes, or temporary barriers. The heatmaps are therefore starting points for visual inquiry, not final ground-truth labels.

3 Applications and Findings

The system enables new forms of urban reading for architects, sociologists, and planners. It can surface cultural gradients by mapping where stylistic markers (e.g., “gothic”) or cultural signals (e.g., “Chinese”) recur, producing neighborhood boundaries from visual density rather than administrative zip codes (Figure 3). It can also make infrastructure legible by indexing the city’s churn: a query for “scaffolding” reveals the temporary city of renovation, while proxies like “air conditioning” suggest building age and retrofit status (Figure 4). Finally, the atlas functions as an everyday lexicon, letting users quantify the visual prevalence of unmapped objects, from subway vents to street art.

Beyond visual exploration, these layers can also serve as a basis for urban studies and digital humanities research. Because each query produces a spatial distribution, the resulting layer can be compared with demographic, economic, environmental, or institutional datasets. Public art can be read against income gradients, redevelopment pressure, or arts districts; cultural markers against commercial corridors and immigration geographies; and civic or infrastructural traces against schools, transit, tourism, or public investment. These comparisons do not make the visual layer causal on its own, but they create a measurable bridge between the visible city and the datasets through which cities are usually studied.



Figure 3: Open-vocabulary neighborhood delineation: the atlas outlines cultural and stylistic regions through repeated visual cues rather than explicit GIS layers.

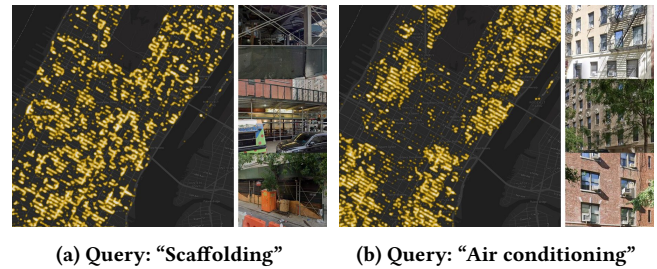


Figure 4: Two readings of infrastructure: (left) the temporary city of construction; (right) building retrofit signals.

4 Physical Installation

The screen interface makes the city searchable; the physical interface makes that search collective. We built a projection-mapped installation around a three-dimensional model of Manhattan, using the model as a surface for semantic heatmaps (Figure 5). When a query such as “murals,” “bridges,” or “skyscrapers” is entered, the result appears both as a browser layer and as a projection across blocks, waterfronts, and corridors.

This format extends prior work on tangible urban interfaces and augmented city models [Ishii et al. 2002; Noyman 2022]. On a screen, search is individual. On the model, the same query becomes visible to a room at once, allowing viewers to point to clusters, compare them with retrieved street-level images, and debate whether the pattern is meaningful, partial, or misleading. The installation turns open-vocabulary search into a shared act of urban interpretation.

5 Challenges and Limitations

These limitations are part of the work’s claim: the atlas is bound by the physics of the street-view camera and the semantics of the model. Occlusion turns objects into absences; lighting collapses detail; and the camera’s chosen routes omit courtyards, rooftops, and interior life. These limitations are not only technical error; they are cartographic arguments. Critical cartography reminds us that maps are made, not found [Harley 1989; Wood 2010]. In open-vocabulary mapping, the “made” quality becomes doubly visible: the map is produced by a chain of translations (world → image → caption → query), each of which can introduce bias.



(a) Projection-mapped model.



(b) Close-up of a projected heatmap.

Figure 5: *Searchable.City* as a physical installation. The same semantic layers used in the browser are projected onto a three-dimensional model of Manhattan.

6 Conclusion

Searchable.City asks a simple question with large consequences: what happens when a map becomes a search engine over images? The answer is a new kind of interface: one where perception itself becomes spatially navigable, and the visual world can be searched, compared, and interpreted through language. The project also exposes the politics of machine vision: the city that AI sees is constrained by where cameras go, what frames exclude, and which words models reach for first. Open-vocabulary cartography does not replace the map of coordinates; it complements it with a map of surfaces. If the twentieth century gave us navigation, the next may give us perception at scale: a city we can query not only for where, but for what.

Project Availability

Searchable.City is publicly available at <https://searchable.city>, where users can explore interactive semantic atlases across multiple cities, including New York City and Los Angeles. The method is city-agnostic: additional regions can be added by ingesting geolocated street-level imagery, extracting open-vocabulary semantic descriptors, and rebuilding the spatial text index. Source code is available at <https://github.com/seanhlewis/searchablecity>.

References

- J. B. Harley. 1989. Deconstructing the Map. *Cartographica*, 26(2), 1–20.
- Hiroshi Ishii, John Underkoffler, D. Chak, Ben Piper, Eran Ben-Joseph, Luke Yeung, and Zahra Kanji. 2002. Augmented Urban Planning Workbench: Overlaying Drawings, Physical Models and Digital Simulation. In *Proceedings of the International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE Computer Society, 203–211. <https://doi.org/10.1109/ISMAR.2002.1115090>
- Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. 2023. Segment Anything. *arXiv:2304.02643 [cs.CV]* <https://arxiv.org/abs/2304.02643>
- Nikhil Naik, Jade Philipoom, Ramesh Raskar, and César A. Hidalgo. 2014. Streetscore – Predicting the Perceived Safety of One Million Streetscapes. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. https://openaccess.thecvf.com/content_cvpr_workshops_2014/W20/papers/Naik_Streetscore_-_Predicting_2014_CVPR_paper.pdf
- Ariel Noyman. 2022. *CityScope: An Urban Modeling and Simulation Platform*. Ph.D. dissertation. Massachusetts Institute of Technology.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. *arXiv:2103.00020 [cs.CV]* <https://arxiv.org/abs/2103.00020>
- Pavan Kumar Anasosalu Vasu, Fartash Faghri, Chun-Liang Li, Cem Koc, Nate True, Albert Antony, Gokul Santhanam, James Gabriel, Peter Gräsch, Oncel Tuzel, and Hadi Pouransari. 2024. FastVLM: Efficient Vision Encoding for Vision Language Models. *arXiv:2412.13303 [cs.CV]* <https://arxiv.org/abs/2412.13303>
- Denis Wood. 2010. *Rethinking the Power of Maps*. Guilford Press.